CS-104 Retrieval Quiz

Fall 2026 · Week 7, Monday

Name: _____

1.

What do the two print statements output?

```
import numpy as np
values = [4, 5]
arr = np.array(values)
print(values * 3)
print(arr * 3)
```

a) `[12, 15]`, then `[12 15]`

b) `[4, 5, 4, 5, 4, 5]`, then `[12 15]`

c) `[4, 5, 4, 5, 4, 5]`, then `[4 5 4 5 4 5]`

d) A `TypeError` for both

2.

Which call produces the array `[0. 0.25 0.5 0.75 1.]`?

a) `np.arange(0, 1, 0.25)`

b) `np.linspace(0, 1, 4)`

c) `np.arange(0, 1, 5)`

d) `np.linspace(0, 1, 5)`

3.

What does this code print?

```
import numpy as np
x = np.array([0, 1, 2, 3])
y = x ** 2
print(y)
```

a) `[0 1 4 9]`

b) `[0 2 4 6]`

c) `[0, 1, 2, 3, 0, 1, 2, 3]`

d) It raises a `TypeError`

4.

A radioactive sample decays over time. The amount left after `t` years is

N = 100 × e^(-0.5 t)

Write a code segment that uses NumPy to create 50 evenly spaced times from 0 to 8 years, computes the amount left at each time (use `np.exp`), and uses Matplotlib to plot amount versus time with the axis labels `Time (years)` and `Amount (g)` and a title.